Life and Death

-Natural Languages vs. Programming Languages-

Composed By: Hayato Ishida

For Japanese version, click here

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About Me

- Accounts
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  - Speaker Deck: @hayat01sh1da
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- Occupation: Software Engineer
- Things I Am Into
  - Linguistics
  - Singing at Karaoke
  - Listening to Music
  - Watching Movies
  - Cat Feeding

Licences / Certifications

- English
  - TOEIC® Listening & Reading 915: Certified on December 2019
- Engineering
  - Information Security Management: Certified on November 2017
  - Applied Information Technology Engineer: Certified on June 2017
  - Fundamental Information Technology Engineer: Certified on November 2016
  - IT Passport: Certified on April 2016
- Others
  - Abacus 2nd Class: Certified on June 2002
  - Mental Arithmetic 3rd Class: Certified on February 2001

Skill Stack

- Languages
  - Japanese: Native Proficiency
  - English: Full professional Proficiency
- Development
  - **Backend: Ruby(Ruby on Rails)**, Python
  - Frontend: TypeScript + React.js, TypeScript +Vue.js
  - **Database: MySQL**, PostgreSQL, MongoDB
  - **Architecture: Monolith, Modular Monolith**
  - Hosting: AWS ESK
  - Versioning: Git, GitHub
  - **CI/CD: GitHub Actions**, ArgoCD
  - Monitoring: Datadog, Sentry

Work Histories

Period	Genre	Position	Jobs
Aug 2021 - Present	Saas(Educational Service)	Software Engineer	• Backend Development • Frontend Development • CI/CD Maintenance • Weekly Release Manager • Troubleshooting • Documentation • Technical Debt Resolution • Tech Blog Entries • LT Speaker
Feb 2020 - Dec 2020	SaaS(Chatbot Platform)	Backend Engineer	• Backend Development • Documentation • Technical Debt Resolution • Inspection of Alternative API

Work Histories

Period	Genre	Position	Jobs
Jun 2018 - Jan 2020	Contract-basis Developer	Software Engineer	• Backend Development • Frontend Development • Troubleshooting • Quality Assurance • Tech Blog Entries
Apr 2016 - Jan 2018	System Engineering Service	System Engineer	• Corporate Account Admin • Windows Server Maintenance • Security Admin Assistant • Internationalisation / Localisation

International-Exchange Activities

- Activities in University
  - English Linguistics Seminar(Focusing on Mass Media English)
  - International-Exchange Clubs(The 2nd Year)
  - International-Exchange Programmes conducted by Japan Cabinet Office(2013 - 2016)
  - Japanese Linguistics Course(The Final Year)
- Overseas Life Experience
  - Working Holiday in Australia(April 2014 - March 2015)
    - Language School for 1 month in Sydney
    - Work for 6 Months in Hamilton Island Resort
    - Volunteering for 1 Month as Assistant Teacher of Japanese Language at St Ives High School in NSW
- Other Activities
  - Keep Everyday Journal in English (April 2014 - Present)
  - Sunrise Toastmasters Club(February 2017 - March 2018)
  - Vital Japan(January 2018 - July 2019, October 2022 - February 2023)
  - Self Learning and Training of English Language
  - Participation Ruby-Related Tech Conferences

The Main Theme

I would love to share the following 3 interesting point of views to compare **Natural Language** with **Programming Language**, whose objectives, purposes and usages are totally different.

- What is the definition of **Live Languages**?
- What is the definition of **Dead Languages**?
- What **a dead language brings about**?

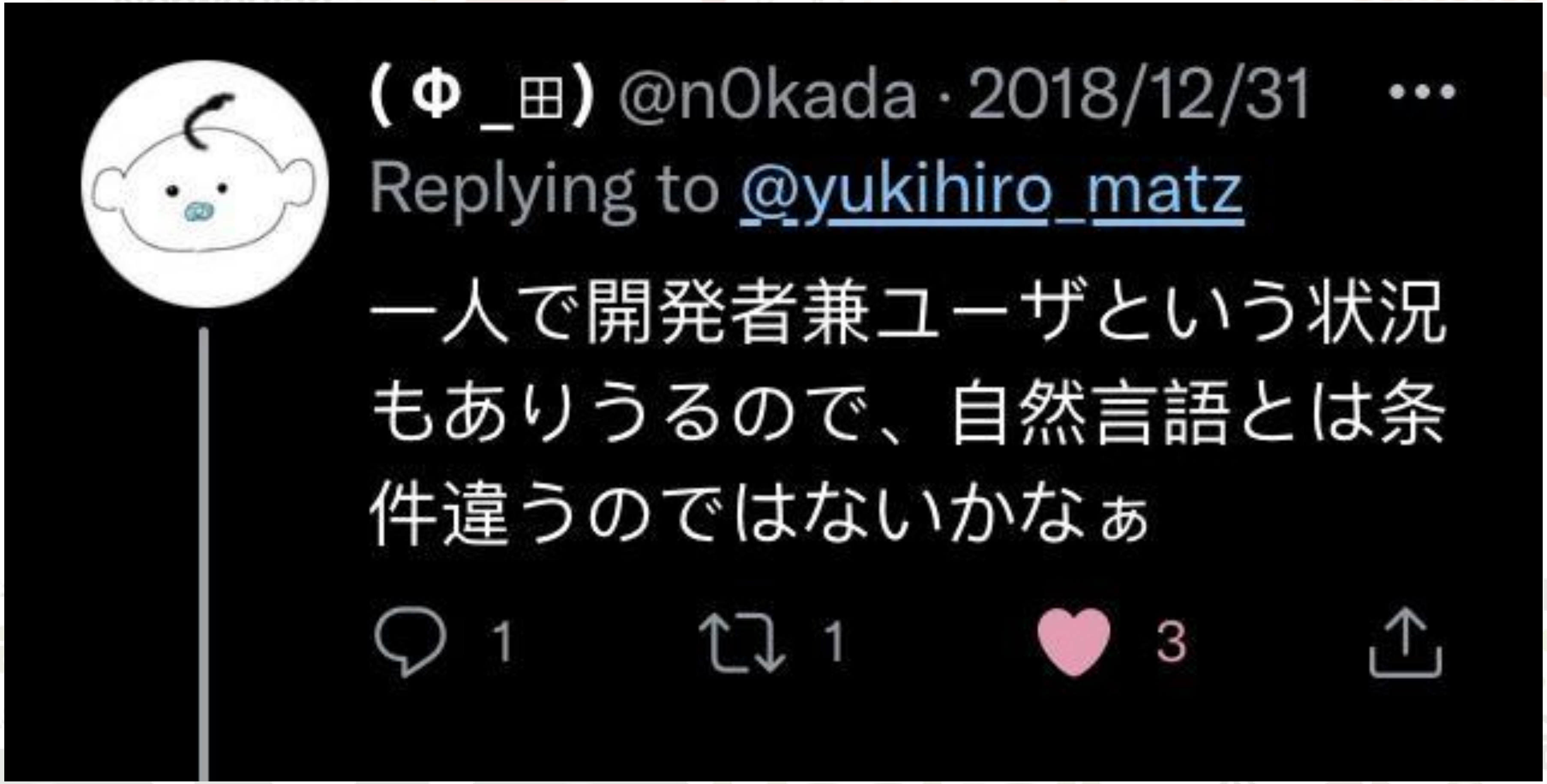
The Theme out of Scope

- The Details of English Language (Words, Idioms, Grammar, Pronunciation etc.)
- The Details of Linguistics(History, Acoustics etc.)
- The Details of Programming Languages(Mechanisms, History, Implementation etc.)

The Original Material

- The following articles as Advent Calendar 2018 in the 2nd Company
- Matz(The Father of Ruby) retweeted them as follows

Discussions Are Welcomed!!



Agenda

1. Representative Features of Each Language
2. Definition of Life and Death
3. Effects Caused by Death
4. Summary
5. References

1. Representative Features of Each Language

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Natural Languages

- Naturally Born
  - Not artificially invented
- Used as both Written and Spoken
  - Formal and Casual
- Living Standard
  - Always changing and evolving
  - Categorised by period like English language as Old English, Middle English, Modern English and Present-day English
- Reflection of Societies, Cultures, Histories and People who Deal with them
- Communication Tool
  - Exchange of information, Regulation of someone's act, Organising knowledge, Inference etc.

1. Representative Features of Each Language

Programming Languages

- Artificial
  - Invented for development by human beings
- Unable to Be Spoken
  - "Do you speak Ruby?" → "Absolutely NO!"
- Constantly Updated and Evolved
  - 3.1.2 of Ruby, 3.10.7 of Python etc.
  - JavaScript is Living Standard(ES6 is a snapshot, not a version)
- Reflection of Developers' Thoughts and Beliefs(including creators and committers)
  - Design of classes and functions
  - Development methods (TDD, DDD etc.)
- Indirect Communication Tool
  - Good codes reflect what intention the developer has
  - The review and reviewee arrive at an agreement via codes which are expected to fulfil requirements

2. Definition of Life and Death

2. Definition of Life and Death

Natural Languages

The true definition of a dead language is one that has no native speakers left. There are several different ways that it can happen, but the bottom line is that if there is only one person left who speaks the language as their native tongue and fluently, then the language has died.

Reference: ll.8-10, Sarah-Claire Jordan, *Alpha Omega Translations*

2. Definition of Life and Death

Natural Languages

- Definition of Life
  - To continue changing and evolving
    - More than or equal to 2 live speakers
    - Communication between native speakers
- Definition of Death
  - No more changes and updates
    - Less than or equal to only 1 speaker
    - No more communication between native speakers
    - We human beings are the alteregoisms of languages

      → If one dies, the other DOES. (Just like the relationship between Piccolo and the God in Dragon Ball Series)

2. Definition of Life and Death

Programming Languages

There are only two kinds of languages: the ones people complain about and the ones nobody uses. - Bjarne Stroustrup, creator of C++.

Be it because nobody is using it, nobody is hiring for it, or nobody is talking about it - based on the level of community engagement, the job market, and overall growth - some languages just aren't worth your time anymore.

Reference: ll.9-15, Justina H., [Codementer]

2. Definition of Life and Death

Programming Languages

- Definition of Life
  - People are concerned about or interested in them
  - Complaints = Demand for them to be more usable
    - Ruby and Ruby on Rails are vigorously alive notwithstanding the yearly myth of death
    - Ruby has traditionally been updated to a major version in every Christmas
  - Developers keep using and enhancing them
- Definition of Death
  - No one pays attention
    - No complaint = No demand
    - No maintenance = Lack in required features and full of security holes
    - **No news is bad news.**

3. Effects Caused by Death

3. Effects Caused by Death

Natural Languages

Orwell believed that if something is not sayable, it will not be thinkable.

In his novel, he told of a society that tried to limit language by getting rid of certain words(e.g., freedom or justice) and restricting the meaning of others.

The purpose was to make certain political ideas unsayable in the hope that they would become unthinkable.

Reference: p.61 ll.17-22, Michael L. Geis, *Language and Communication*

3. Effects Caused by Death

Natural Languages

- Loss of the Way to Think
  - How do you think about and express *hunger* without the word
  - If a nation or society regulates the language, it means to force the people to lose the way they think about and articulate specific thoughts
- Loss of Cultural Identities
  - Honorifics and euphemisms specific in Japanese language have been molded in the long history of Japan, for example
  - They have been thought, expressed and developed via loads of use via the language
  - It will absolutely be impossible for the Japanese specific concepts such as its two-faced culture and pleasantries to stay present

3. Effects Caused by Death

Programming Languages

- Falls of the Market Values
  - "I developed i-Mode in Doja for a long time and have a high expertise of it" no more works
    - * i-Mode is a domestics web service for cellphones which used to be provided by Docomo
    - However, nothing is accurately predictable because JavaScript revived thanks to Ajax for Google Map using asynchronous communications after it had been swallowed by the high trend of Flash
- Increments of Development Costs
  - Maintenance costs
  - Implementation of lacking features from the scratch
  - Security patches
  - Transfer to another programming language or framework
  - etc.

4. Summary

4. Summary

Nature	Natural Languages	Programming Languages
Natural / Artificial	Naturally Born	人工的
Written / Spoken	Written and Spoken	Written Only
Change and Evolution	Always	Constantly(It depends)
Reflection	Individuals, Communities, Societies, Cultures, Histories etc.	Thoughts and Belief of Developers (Creators and Committers)
Communication Tool?	○	△
Definition of Life	To Continue Changing and Evolving	For People to Stay Interested
Definition of Death	Less than or Equal to 1 Native Speaker → No More Communication → No More Changes and Updates	None of anyone's business → No More Development → No More Changes and Updates
Effects Caused by Death	Loss of the Way to Think Loss of Cultural Identities	Falls of Market Values Increments of Development Costs

4. Summary

Supplementary Comments

- On the one hand, it was difficult to find reliable references for programming languages because few people care about them unless they are highly evaluated and mentioned
- On the other hand, it was much easier to find reliable references for natural languages because linguistics is established as an academic subject for them
- Programming languages are born and raised in the capitalism era while natural languages have way long histories, so they die of the poor market values, not of their language natures
- If *Programming Linguistics* is present…, what would it be like?

5. References

5. References

- Sarah-Claire Jordan, *What Causes the Death of a Language?*
  - Last Accessed On: 23 October 2021
- Justina H., *Worst Programming Languages to Learn in 2018*
  - Last Accessed On: 23 October 2021
- Michael L. Geis, *Language and Communication*, Oxford, OUP, 2001
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  - Last Accessed On: 09 August 2021
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